

# Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

**Benchmark:** ARABICMMLU | **Context:** MENA Multi-Country Arabic Tutoring System — Eight-Country Curriculum Deployment

**Input Content definition:** Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

Content-level validity is the dimension where ArabicMMLU diverges most sharply from the deployment's eight-country requirement. While the benchmark's content is sourced from real native Arabic exams [Q1, Q26] — a clear strength over translated alternatives [Q5, Q6] — the empirical content distribution is heavily skewed: civics, social science, history, and economics at school level are dominated by Jordanian-specific institutional, royal, and constitutional content presented as universal curriculum knowledge [DATASET-D1, D2, D3, D4, D5, D6, D7, D20, D21, D22, D25, D30, D31]. University-level economics, accounting, management, and political science come exclusively from a single Egyptian university (Assiut University) in the sample [DATASET-D13, D14, D28, D39, D40], and Law (Professional) is exclusively Moroccan in the observed sample [DATASET-D8, D9, D10, D34]. The deployment user explicitly requires country-specific tailoring for law [elicitation A2] and curriculum-aligned content for each of the eight countries [elicitation A1]. Kuwait has no observed examples and no documented annotators [Q31, Q96]; Morocco has no documented annotators [Q31, Q96]. Egypt-specific currency (ج.م.ع) appears in university economics presented as universal [DATASET-D28]. The benchmark's own admission that countries are represented unequally [Q89] is empirically confirmed and amplified by the dataset analysis: rather than uniform under-coverage, the data shows Jordanian-curriculum content is propagated as default for non-Jordanian cohorts. This generates substantial construct-irrelevant variance for Moroccan, Emirati, Kuwaiti, Saudi, Lebanese, and Palestinian students.

| ID  | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| Q1  | "we present ArabicMMLU, the first multi-task language understanding benchmark for the Arabic language, sourced from school exams across diverse educational levels in different countries spanning North Africa, the Levant, and the Gulf regions." (p.1)                                                                                                                                                                                                                              |
| Q5  | "Although large language models (LLMs) such as GPT-3.5 (Ouyang et al., 2022), BLOOMZ (Muennighoff et al., 2022), and Jais (Sengupta et al., 2023) have been pretrained with substantial coverage of Modern Standard Arabic (MSA), their reasoning and knowledge assessments are primarily conducted using datasets translated from English to Arabic (Sengupta et al., 2023; Huang et al., 2023), which means there is limited capacity to evaluate content specific to Arabic." (p.1) |
| Q6  | "This reliance on translation systems not only demonstrates an Anglocentric approach (Ramesh et al., 2023; Talat et al., 2022) but also potentially introduces errors and biases." (p.1)                                                                                                                                                                                                                                                                                               |
| Q26 | "The questions are sourced from eight different countries in North Africa (Morocco and Egypt), the Levant (Jordan, Palestine, and Lebanon), and the Gulf (UAE, Kuwait, and KSA)." (p.3)                                                                                                                                                                                                                                                                                                |
| Q31 | "The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4)                                                                                                                                                                                                                                |
| Q89 | "ArabicMMLU does not represent all Arabic countries equally. For example, we have collected over 6K multiple-choice questions from Jordan, while other countries are represented with only 100 questions or, in some cases, not at all." (p.9)                                                                                                                                                                                                                                         |
| Q96 | "We extend our gratitude to all collaborators from Jordan, Egypt, Lebanon, UAE, and Saudi Arabia who participated in the data collection process." (p.10)                                                                                                                                                                                                                                                                                                                              |
| D1  | What are the manifestations of change in modern Jordanian society? — country-specific Jordanian civics                                                                                                                                                                                                                                                                                                                                                                                 |
| D2  | The first coalition government in Jordan in 1952 was known as the ___ government — Jordanian political history                                                                                                                                                                                                                                                                                                                                                                         |
| D3  | The council with general sovereignty over all state affairs is the Council of Ministers — specifically Jordanian constitutional structure                                                                                                                                                                                                                                                                                                                                              |
| D4  | Rights under the [Jordanian] Kingdom's constitution — explicitly references the Jordanian Hashemite Kingdom                                                                                                                                                                                                                                                                                                                                                                            |

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| D5  | "لكلمة الإلحاح دعه في 1952 ماع روتسد رص" with options referring to Jordanian kings — "The 1952 constitution was issued during the reign of King [Talal bin Abdullah]" — Jordanian royal history                                                                            |
| D6  | وهو يزكركم كنبلل ينورتكلال عقومل قيرط نع يوكشلال لاسرا www.cbj.gov.jo — "Complaints to the Central Bank [of Jordan] via www.cbj.gov.jo" — explicitly references Jordan's central bank                                                                                      |
| D7  | 0.67: نسبة من الأردنيين لا يستطيعون الوصول إلى الخدمات المالية الرسمية — "67% of Jordanians cannot access formal financial services" — Jordan-specific statistic as universal economic fact                                                                                |
| D8  | — يوعدل رهوج في مكحل رودص دعب الإ عوفد وأ ضراعازن في مرداصل وأ عيدي مهممتل ماكحلأ فائتتسا لبقي ال Moroccan procedural law — appeals rules under Moroccan Code of Civil Procedure                                                                                           |
| D9  | عافدل اب فلكملم عيوكحل اطلسل او لدعل ربزو نم رارب...نيكي كدلل عيئاضلق اطرشل طباض ففص لي وخت نكمي "يكللمل كدلل" (Royal Gendarmerie of Morocco) — Moroccan judicial police powers — references                                                                               |
| D10 | — Moroccan civil party withdrawal in criminal proceedings — عحي حص بوقلأ عي مج: مكحل رودص لبقي ندمل فرطل لزانن اذإ                                                                                                                                                         |
| D13 | — عي عبتال ب عي مس نوكت لي غشتل ا عي ممع نإف... عي مس تال خدمل تنك اذإ ونأ بساحل لوح ععجارمل بومسأ ضرقتفي Computer auditing concept — general accounting principle, from Egyptian source                                                                                   |
| D14 | Severely — لي ري تذت... فتتلفل خسل ا تي رادل ا تاي هتتدسل او صاختل ألي ني تب عطلتدل عتي هت فتجرد OCR-distorted text about authority distribution in organizations, from Egyptian source                                                                                    |
| D20 | "Royal Medical Services include: Medical City Hospital" — عي بطل ا ن عي دمل ي فشتسم: اهنم... تاسس ؤمل نم دي دعل ا عي وكحل ا عي كمللم ا عي بطل ا تادمخل ا لمشت Royal Medical Services specifically Jordanian Royal Medical Services                                         |
| D21 | "King Talal bin Hussein's reign ended for health reasons in 1952 — one year" — عي كذو عي حص بابسأل هتالو تهتن ا ن عي سحل نب لال طلكلم King Talal bin Hussein's reign ended for health reasons in 1952 — one year — Jordanian royal history in primary school               |
| D22 | — Lady of the Mountain Church is in Anjara [Jordan] — عي ل بجل ا عي سي نك دجوت Jordanian religious geography                                                                                                                                                               |
| D25 | "The main goal of the 1928 Jordanian-British treaty was liberation from Mandate restrictions" — عي ناطي ربل ا هني درأل ا هدا عمل سيئرل ا فدهل ناك The main goal of the 1928 Jordanian-British treaty was liberation from Mandate restrictions — Jordanian national history |
| D28 | — Egyptian currency (ج = ١٠٠٠٠٠ بولطم رامثتسال مجج ناك اذإ ١٠ % : وه عرمثتسمل ل او مال ا ل ع دئاعل ا لدعم... هني ج ١٠٠٠٠٠ بولطم رامثتسال مجج ناك اذإ Egyptian pound) in university economics — Egypt-specific monetary context                                             |
| D30 | "Wadi Rum is a Jordanian nature reserve under joint management" — مري داو... كعرتشم اراذل ا هيل ا فرشت ندرأل ا في عي عي بط عي مجج Wadi Rum is a Jordanian nature reserve under joint management — Jordan-specific geography fact                                           |
| D31 | King Abdullah I's positions on Zionism — عي ناطسلف في اعا ماطأ و عي نو يهصل ا كرحل ا نم لوال ا هل لادبع لكلمل ا فقاوم نم تسيل ا عي تال ا نم دحاو King Abdullah I's positions on Zionism — highly sensitive political content, Jordanian framing of Palestinian history     |
| D34 | All 5 Law (Professional) examples: Country = Morocco, from single Google Drive PDF — All sampled law questions are exclusively from Morocco — single-source PDF                                                                                                            |
| D39 | Source: http://www.aun.edu.eg/commerce/... — All 5 accounting examples from Assiut University (Egypt) — single Egyptian university source                                                                                                                                  |
| D40 | Source: http://www.aun.edu.eg — all 5 from same Assiut University Egypt URL — University economics exclusively from Egyptian university source                                                                                                                             |

## Strengths

- Native exam provenance avoids translation artifacts that plague competing Arabic benchmarks [Q5, Q6]
- STEM content (math, biology, physics, CS factual) is genuinely curriculum-neutral and reliable across all eight countries [DATASET-D24]
- Palestine is substantively represented in STEM subjects (Biology HS, CS Primary, Math Primary) [DATASET-D23, D24]
- Some Moroccan legal content is present, even if narrow [DATASET-D8, D9, D10]
- Native annotators and exam URLs document content provenance per question [Q32, Q33]

| ID | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| Q5 | "Although large language models (LLMs) such as GPT-3.5 (Ouyang et al., 2022), BLOOMZ (Muennighoff et al., 2022), and Jais (Sengupta et al., 2023) have been pretrained with substantial coverage of Modern Standard Arabic (MSA), their reasoning and knowledge assessments are primarily conducted using datasets translated from English to Arabic (Sengupta et al., 2023; Huang et al., 2023), which means there is limited capacity to evaluate content specific to Arabic." (p.1) |
| Q6 | "This reliance on translation systems not only demonstrates an Anglocentric approach (Ramesh et al., 2023; Talat et al., 2022) but also potentially introduces errors and biases." (p.1)                                                                                                                                                                                                                                                                                               |

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| <b>Q32</b> | "During the first stage of data collection, the internal workers were tasked with collecting relevant sources for data collection. These sources were URLs containing the questions, which needed to be publicly available." (p.4)                                                    |
| <b>Q33</b> | "In the second stage, all workers were asked to manually scrape the data within a 2-month period. The task was to collect metadata, including the source (URL of the source document), country, subject, level, question, multiple-choice options, and the correct answer key." (p.4) |
| <b>D8</b>  | — Moroccan procedural law — appeals rules under Moroccan Code of Civil Procedure<br>يؤعدل رهوج يف مكحل رودص دعب الإ عوفد وأ ضراع عازن يف ةرداصل وأ ةيديهمم الت ماكحلأ فانئتسا لبقي ال                                                                                                 |
| <b>D9</b>  | عافدل اب ةفل كم لم ةي موكحل اطلسل او لدعل ربزو نم رارقب...نني كيردل ل ةيئاضق ل طرشل طباض ةفص لي وخت نكمي<br>Moroccan judicial police powers — references "لكم ل كيردل" (Royal Gendarmerie of Morocco) — ينطولا                                                                        |
| <b>D10</b> | Moroccan civil party withdrawal in criminal proceedings — عحي حص ةبوجلأ عي م ح: مكحل رودص لبقي ندمل فرطل ل زانت اذا                                                                                                                                                                   |
| <b>D23</b> | "The binary number system is used by computers" — Palestine-sourced primary CS<br>دأ — ةيئانثل ماظنل: اهتجل اعمو تاناي بل طفحو لي ثمت يف بوساحل هم دختسي دعل ةمظنأ دح                                                                                                                 |
| <b>D24</b> | Calvin cycle question — Palestine-sourced biology; universal factual content<br>إذا — Calvin cycle question — زوكولغ تائي زج (3) جتن اذا                                                                                                                                              |

## Checklist

### Checklist item definitions:

| ID          | Assessment Question                                                        |
|-------------|----------------------------------------------------------------------------|
| <b>IC-1</b> | Do inputs require region-specific cultural/geographic/dialectal knowledge? |
| <b>IC-2</b> | Does culturally sensitive content align with target culture?               |
| <b>IC-3</b> | Flag inputs assuming Western-specific knowledge                            |
| <b>IC-4</b> | Regional annotator recruitment for culturally sensitive instances          |
| <b>IC-5</b> | Document content issues harming content validity                           |

### Checklist responses:

**IC-1:** Inputs heavily require region-specific knowledge but the regions covered diverge from the deployment's eight-country target. Civics and history items reference Jordanian royal lineage, the 1952 Jordanian constitution, Jordan's Royal Medical Services, and Central Bank of Jordan URLs as universal facts [DATASET-D1, D2, D4, D5, D6, D20, D21, D25, D31]. University accounting and economics use Egyptian institutional and currency context [DATASET-D28, D39, D40]. Law uses Moroccan procedural code [DATASET-D8, D9, D10].

**IC-2:** Cultural-content alignment is partial. Islamic studies content observed (Hadith ethics, Quran recitation etiquette) appears broadly Sunni-consistent, aligning with the deployment's predominantly Sunni population [DATASET-D17, D37, D38]. However, no madhab calibration exists [gap\_id 7], and Hanbali/Maliki specificity is not verified. Politically sensitive content (e.g., King Abdullah I and Zionism [DATASET-D31]) is framed in Jordanian national narrative — potentially misaligned with Palestinian, Lebanese, or other framings.

**IC-3:** Western-specific knowledge is largely avoided due to native exam sourcing [Q5, Q6]. The primary content-validity threat is not Western bias but intra-regional bias — Jordanian and Egyptian content dominating non-Jordanian/non-Egyptian cohorts.

**IC-4:** Annotators are exclusively from Jordan, Egypt, Lebanon, UAE, KSA [Q31] — Morocco, Kuwait, Palestine have no documented annotators despite (in Palestine's case) contributing top-three question volume [Q96, WEB-1]. Recruiting Kuwaiti, Moroccan, and Palestinian curriculum experts would be essential for any deployment use.

**IC-5:** Documented content issues harming validity: (a) Jordanian-default civics/history/social science presented as universal [DATASET-D1, D2, D4, D21]; (b) single-Egyptian-university dominance of university social science subjects [DATASET-D39, D40]; (c) single-source single-jurisdiction Law (Professional) [DATASET-D34]; (d) no Kuwait content observed; (e) Egypt-specific currency in 'general' economics [DATASET-D28]; (f) French-medium Moroccan university content absent [Q92].

| ID | Evidence |
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| Q5    | "Although large language models (LLMs) such as GPT-3.5 (Ouyang et al., 2022), BLOOMZ (Muennighoff et al., 2022), and Jais (Sengupta et al., 2023) have been pretrained with substantial coverage of Modern Standard Arabic (MSA), their reasoning and knowledge assessments are primarily conducted using datasets translated from English to Arabic (Sengupta et al., 2023; Huang et al., 2023), which means there is limited capacity to evaluate content specific to Arabic." (p.1) |
| Q6    | "This reliance on translation systems not only demonstrates an Anglocentric approach (Ramesh et al., 2023; Talat et al., 2022) but also potentially introduces errors and biases." (p.1)                                                                                                                                                                                                                                                                                               |
| Q31   | "The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4)                                                                                                                                                                                                                                |
| Q89   | "ArabicMMLU does not represent all Arabic countries equally. For example, we have collected over 6K multiple-choice questions from Jordan, while other countries are represented with only 100 questions or, in some cases, not at all." (p.9)                                                                                                                                                                                                                                         |
| Q92   | "The dataset primarily focuses on Modern Standard Arabic (MSA). However, multilingual and Arabic LLMs are often exposed to both MSA and dialectal Arabic." (p.9)                                                                                                                                                                                                                                                                                                                       |
| Q96   | "We extend our gratitude to all collaborators from Jordan, Egypt, Lebanon, UAE, and Saudi Arabia who participated in the data collection process." (p.10)                                                                                                                                                                                                                                                                                                                              |
| D1    | "What are the manifestations of change in modern Jordanian society?" — country-specific Jordanian civics                                                                                                                                                                                                                                                                                                                                                                               |
| D2    | "The first coalition government in Jordan in 1952 was known as the ___ government" — Jordanian political history                                                                                                                                                                                                                                                                                                                                                                       |
| D4    | "Rights under the [Jordanian] Kingdom's constitution" — explicitly references the Jordanian Hashemite Kingdom                                                                                                                                                                                                                                                                                                                                                                          |
| D5    | "The 1952 constitution was issued during the reign of King [Talal bin Abdullah]" — Jordanian royal history                                                                                                                                                                                                                                                                                                                                                                             |
| D6    | www.cbj.gov.jo — "Complaints to the Central Bank [of Jordan] via www.cbj.gov.jo" — explicitly references Jordan's central bank                                                                                                                                                                                                                                                                                                                                                         |
| D8    | appeals rules under Moroccan Code of Civil Procedure                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| D9    | references "Royal Gendarmerie of Morocco" (Moroccan judicial police powers)                                                                                                                                                                                                                                                                                                                                                                                                            |
| D10   | Moroccan civil party withdrawal in criminal proceedings                                                                                                                                                                                                                                                                                                                                                                                                                                |
| D17   | Country: Jordan — 5-option Islamic studies question about positive effects of thinking                                                                                                                                                                                                                                                                                                                                                                                                 |
| D20   | "Royal Medical Services include: Medical City Hospital" — specifically Jordanian Royal Medical Services                                                                                                                                                                                                                                                                                                                                                                                |
| D21   | "King Talal bin Hussein's reign ended for health reasons in 1952 — one year" — Jordanian royal history in primary school                                                                                                                                                                                                                                                                                                                                                               |
| D25   | "The main goal of the 1928 Jordanian-British treaty was liberation from Mandate restrictions" — Jordanian national history                                                                                                                                                                                                                                                                                                                                                             |
| D28   | Egyptian currency (Egyptian pound) in university economics — Egypt-specific monetary context                                                                                                                                                                                                                                                                                                                                                                                           |
| D31   | King Abdullah I's positions on Zionism — highly sensitive political content, Jordanian framing of Palestinian history                                                                                                                                                                                                                                                                                                                                                                  |
| D34   | All 5 Law (Professional) examples: Country = Morocco, from single Google Drive PDF — All sampled law questions are exclusively from Morocco — single-source PDF                                                                                                                                                                                                                                                                                                                        |
| D37   | Country: Jordan — Generic Islamic etiquette of Quran recitation — no madhab specificity indicated                                                                                                                                                                                                                                                                                                                                                                                      |
| D38   | Country: Jordan — Islamic studies question about examples in Quran — Jordanian curriculum framing                                                                                                                                                                                                                                                                                                                                                                                      |
| D39   | Source: http://www.aun.edu.eg/commerce/... — All 5 accounting examples from Assiut University (Egypt) — single Egyptian university source                                                                                                                                                                                                                                                                                                                                              |
| D40   | Source: http://www.aun.edu.eg — all 5 from same Assiut University Egypt URL — University economics exclusively from Egyptian university source                                                                                                                                                                                                                                                                                                                                         |
| WEB-1 | Models perform notably poorly on Morocco- and UAE-sourced questions, consistent with thin annotation coverage (https://arxiv.org/html/2402.12840v1)                                                                                                                                                                                                                                                                                                                                    |

Information Gaps

- Whether Jordanian dominance in civics/history/social science extends to the full benchmark or is concentrated in specific level/subject configurations [gap\_id 5]
- Whether Law subset beyond the sampled Moroccan questions includes other jurisdictions [gap\_id 4]
- Madhab-specific framing in Islamic studies is undocumented [gap\_id 7]

Requires Expert Verification

- Per-country curriculum review by national-curriculum experts from Kuwait, Morocco, Palestine, UAE to identify questions presented as 'universal' that are actually national-curriculum-specific
- Islamic studies curriculum experts to assess madhab calibration risk for Moroccan Maliki and Saudi Hanbali contexts

Remediation

**Gap 1: Jordanian-curriculum civics, history, social science, and high-school economics content is encoded as universal across configurations, creating construct-irrelevant variance for the seven non-Jordanian deployment cohorts [DATASET-D1, D2, D6, D7, D21, D25].**

**Recommendation:** Use the per-question country metadata field [Q33] to filter or stratify evaluation by source country; do not aggregate across countries for civics, history, or social science. Build a Kuwaiti, Moroccan, and Palestinian curriculum supplement before deploying to those cohorts.

**Gap 2: Islamic studies has no madhab labeling anywhere in the benchmark [gap\_id 7]; observed Islamic Studies HS content is Jordanian-curriculum-sourced [DATASET-D38] without Maliki (Morocco) or Hanbali (KSA) calibration.**

**Recommendation:** Engage Islamic studies curriculum experts from Morocco (Maliki), KSA (Hanbali), and a Shafi'i-region representative to audit a stratified sample of Islamic studies items and tag madhab-specific items; flag madhab-divergent questions for country-conditioned evaluation.

| ID  | Evidence                                                                                                                                                                                                                                                                              |
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| Q33 | "In the second stage, all workers were asked to manually scrape the data within a 2-month period. The task was to collect metadata, including the source (URL of the source document), country, subject, level, question, multiple-choice options, and the correct answer key." (p.4) |
| D1  | "What are the manifestations of change in modern Jordanian society?" — country-specific Jordanian civics                                                                                                                                                                              |
| D2  | "The first coalition government in Jordan in 1952 was known as the ___ government" — Jordanian political history                                                                                                                                                                      |
| D6  | www.cbj.gov.jo — "Complaints to the Central Bank [of Jordan] via www.cbj.gov.jo" — explicitly references Jordan's central bank                                                                                                                                                        |
| D7  | 0.67 — "67% of Jordanians cannot access formal financial services" — Jordan-specific statistic as universal economic fact                                                                                                                                                             |
| D21 | "King Talal bin Hussein's reign ended for health reasons in 1952 — one year" — Jordanian royal history in primary school                                                                                                                                                              |
| D25 | "The main goal of the 1928 Jordanian-British treaty was liberation from Mandate restrictions" — Jordanian national history                                                                                                                                                            |
| D38 | "Country: Jordan — Islamic studies question about examples in Quran — Jordanian curriculum framing                                                                                                                                                                                    |